Lab 6: Sichuan Mahjong

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Introduction

Mahjong is a tile-based game that was developed in the 19th century in China and has spread throughout the world since the early 20th century. It is commonly played by four players on a square table. Recently, the game has also been adapted into widespread online entertainment.



A Mahjong parlor in Chengdu



An online Riichi Mahjong platform

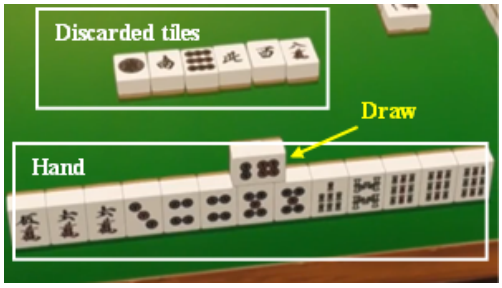
In this lab, you are going to write a program (more precisely, a function) that calculates the score of a Mahjong hand. You can use C/C++ to complete the task. The rules we set up are adapted from Sichuan Mahjong.

Sichuan Mahjong Basic

Sichuan Mahjong consists of 27 different tiles. It's different from some other Mahjong rules that Honor tiles (Wind and Dragon) are counted. For each tile, there are 4 of them, so there are a total of 108 tiles. They belong to 3 suits: **Characters**, **Dots**, and **Bamboos**.

Suits \ Number	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Characters	一萬	二萬	三萬	四萬	伍萬	六萬	七萬	八萬	九萬
Bamboos	1索	2索	3索	4索	5索	6索	7索	8索	9索
Dots	1筒	2筒	3筒	4筒	5筒	6筒	7筒	8筒	9筒

The objective of Mahjong is to win, of course. So how do we win a game? In a game, each player usually has 13 tiles in hand. In every turn, you draw a tile, so the number of tiles becomes 14. Then, you shall discard one from your hand. As you keep drawing and discarding, your hand also changes.

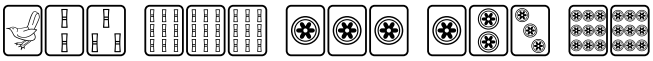
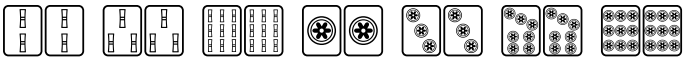


A demonstration of an ongoing round

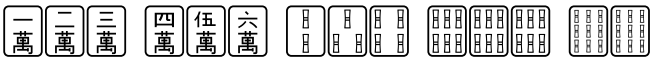
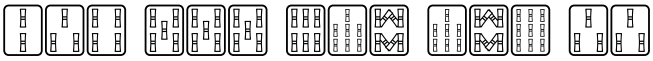
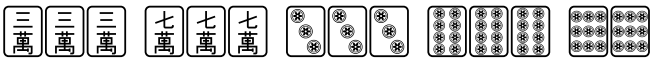
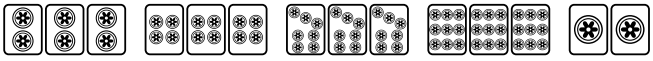
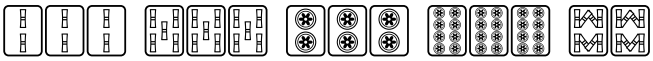
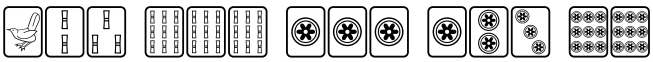
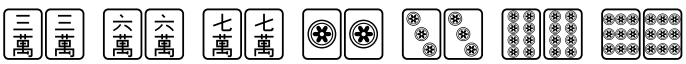
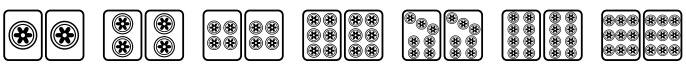
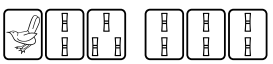
Before explaining how to win, we must first understand three terms: **Sequence**, **Triplet**, and **Pair**.

- A **Pair** is made of 2 identical tiles. e.g. 
- A **Triplet** is made of 3 identical tiles. e.g. 
- A **Sequence** is made of 3 consecutive tiles of the same suit. e.g. 

Usually, when your 14-tile hand consists of "4 **Meld**s (either **Sequence** or **Triplet**) + 1 **Pair**", or consists of "7 **Pair**s", the hand is called a Winning Hand. Of course, remember that in Sichuan Mahjong, players should **pick an unwanted suit** each round. In our rules, we adapt this convention by limiting the suits of tiles the hand contains to at most 2 for a necessary condition to win.

Winning Hand	***Example***
4 **Meld**s + 1 **Pair**	
7 **Pair**s	

Generally speaking, winning more rounds doesn't make you the winner of a game. That's because a Winning Hand varies in score according to its pattern. Every pattern is bound with a *Fann* value, and the score of a Winning Hand is just the *Fann* (multiply by *Bonus*) of its pattern. In this lab, You only need to consider the following 10 situations.

Pattern	***Faan***	**Description**	***Example***
Chicken Hand	1	Not satisfy any other pattern (except *Bonus*).	
Full Flush	4	All tiles are of the same suit.	
All Triplets	2	All melds are **Triplet**s.	
Pure Triplets	8	A *Full Flush* of *All Triplets*.	
2/5/8 Triplets	8	All tiles are numbered 2, 5, or 8.	
Terminals in All Sets	4	All sets contain terminals*.	
Pure Terminals in All Sets	16	A *Full Flush* of *Terminals in All Sets*.	
*Seven Pairs***	4	7 sets of **Pair**s. Same **Pair**s are allowed.	
*Pure Seven Pairs***	16	A *Full Flush* of *Seven Pairs*.	
***Bonus**: Root*	× 2	Have 4 identical tiles. Can be accumulated.	 (the *Root* here is )

- * Namely, the ends of each suit: 🀇, 🀏, 🀐, 🀙, 🀡, 🀘.
- ** If the type of a winning hand is 7 **Pair**s, it will only be counted as either *Seven Pairs* or *Pure Seven Pairs*. Bonus is also applicable for 7 **Pair**s. (Sorry for the difference from real-world Sichuan Mahjong.)

Notice that in Sichuan Mahjong, a Winning Hand should **have and only have one pattern (except *Bonus*)**. If a winning hand satisfies a pattern of higher level (with more *faan* value), the one of lower level will not be counted. For example, A *2/5/8 Triplets* must be an *All Triplets*, but it will only be counted as *2/5/8 Triplets*.

Rarely (but will exist in the testcase), a Winning Hand can be interpreted as different kinds of pattern. If that is the case, only the pattern with more *faan* value with *Bonus* should be counted. For example, the winning hand 🀁🀁🀂🀂🀃🀃🀆🀆🀆🀆🀆🀆🀆🀆 can be interpreted as either *Chicken Hand* (1 *faan* × 2) or *Seven Pairs* (4 *faan* × 2), but only the latter one is counted.

Codename of Hands

To simplify the notation of Mahjong tiles, we adapt a codename system. In the system, There is a unique codename for every single tile, comprising two characters: a digit and a lowercase letter. The digit is the ordinal number of the tile, and the letter represents the suit of that tile, where `c` is for **Character**, `d` is for **Dot**, and `b` is for **Bamboo**. For example, `8d` stands for 🀘, `5c` stands for 🀋, and `1b` stands for 🀐.

The codename of a hand is just a concatenation of the codenames of tiles with no space and other characters in it. For example, the hand of 🀇🀈🀉🀊🀋🀌🀙🀙🀙🀞🀞🀞🀞🀞 can be represented by `1c2c3c4c5c6c2b3b4b6b6b6b9b9b`. Moreover, to make it more concise and convenient to read, the letter between tiles of the same suit can be omitted. i.e. `123456c23466699b` is also a valid codename. Please notice that the codenames are not necessarily listed in order.

Task

Write a function `int getScore(const char strHand[])` in a file named "lab6.c" (or "lab6.cpp" depending on the language you use). The return type, function name, and parameter type **should be unchanged**. The **parameter** `strHand[]` is a C-style string of the codename of a hand of exactly 14 tiles. You don't need to consider invalid inputs like a hand of 13 tiles or a tile with five existences. The **return** of the function should be the score of that hand according to the rules introduced above. (If it is not a Winning Hand, return `0`).

Here is some In/Out example of `getScore()`, more sample cases will be given in another file.

```
In  [0]:  1d2d3d4d5d6d7d8d9d3c4c4c4c5c
Out [0]:  1         // Chicken Hand: 1
In  [1]:  123123123b55588c
Out [1]:  2         // All Triplets: 2
In  [2]:  99987654321111d
Out [2]:  8         // Full Flush + 1 Root: 4 * 2 = 8
In  [3]:  55551166663333d
Out [3]:  128       // Pure Seven Pairs + 3 Roots: 16 * 2 * 2 * 2 = 128
In  [4]:  123777c222b11123d
Out [4]:  0         // Not winning because it contains 3 suits
In  [5]:  12334777c224555d
Out [5]:  0         // Not winning either
```

Please do not write an `int main()` in "lab6.c" (or "lab6.cpp"). The proper way to debug your function is to write another executable file that loads the "lab6.c" (or "lab6.cpp"). There are many ways to do it. One simplest way is, for example, to create a file "main.c" with content as follows.

```
#include <stdio.h>
#include "lab6.c"     // allows you to call function getScore() in this file.

int main() {          // EFFECT: print the score of a hand given through stdin
    char strHand[30];
    scanf("%s", strHand);
    int score = getScore(strHand);
    printf("%d", score);
    return 0;
}
```

Writing a piece of code like this is how we usually do in a real-world collaborating project. It is not necessarily compilable individually, but it performs like a "part" in a large project. In this lab, You can imagine you are contributing to a project of an online Mahjong video game.

Tips

- You may write several other functions and define some structure/class to help you with the task.
- As far as I know, there is not much difference in using C or C++ for this lab, and there is no need to include any library to complete the task.
- Carefully design your algorithm. Algorithm is the most difficult part of this lab. Don't feel frustrated if you cannot come up with a nice solution on your own. You can search on the Internet for ideas. There are plenty of established strategies you can refer to.

Submission

There will be a JOJ test for this lab. Please submit your code according to the instructions on JOJ.

After the lab, remember to compress your file(s) into a single zip file in the following name format: lab6-[Your Last Name]-[Your First Name]-[Your SJTU ID].zip, for example, lab5-Zhong-Zhiyao-521370910153.zip and submit it to the corresponding assignment page on canvas.

Rubric

The full mark of this lab is 140 pts, of which 105 pts are automatically generated by JOJ. They are roughly divided as follows.

- 35 pts - Lab Attendance & Participation.
- 70 pts - Able to tell winning hands and non-winning hands apart. (return >0 for winning hands and 0 for non-winning)
- 35 pts - Able to calculate the score correctly.

Reference

1. Xinran Wang, VG101 FA2018 – Lab 7
2. Wikipedia: Mahjong. https://en.m.wikipedia.org/wiki/Mahjong
3. Mahjong Soul. https://mahjongsoul.yo-star.com